
TCX2005: Information Systems, Management and Organisations

L1: Information Systems in Global Business Today

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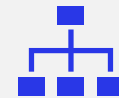
Lesson Objectives



What is an information system?



How does it work and its key stakeholders?



What are the core dimensions of an IS?

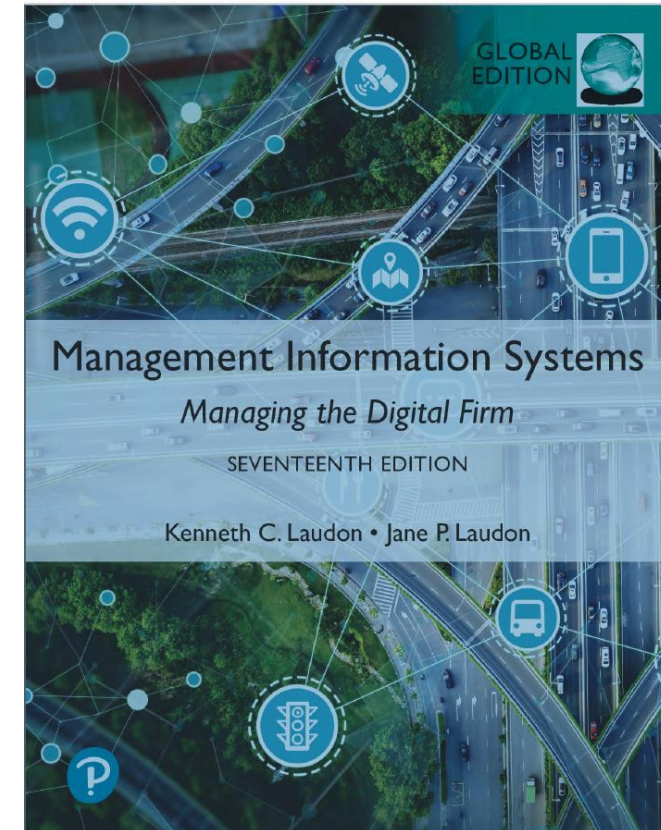


What academic disciplines are used to study information systems and how does each contribute to an understanding of information systems?

Learning Resources



- Textbook (read before class)
 - Read chapter 1 and 2
- Any Other Material
 - Shared on Canvas for this week
 - Videos: [AWM](#),

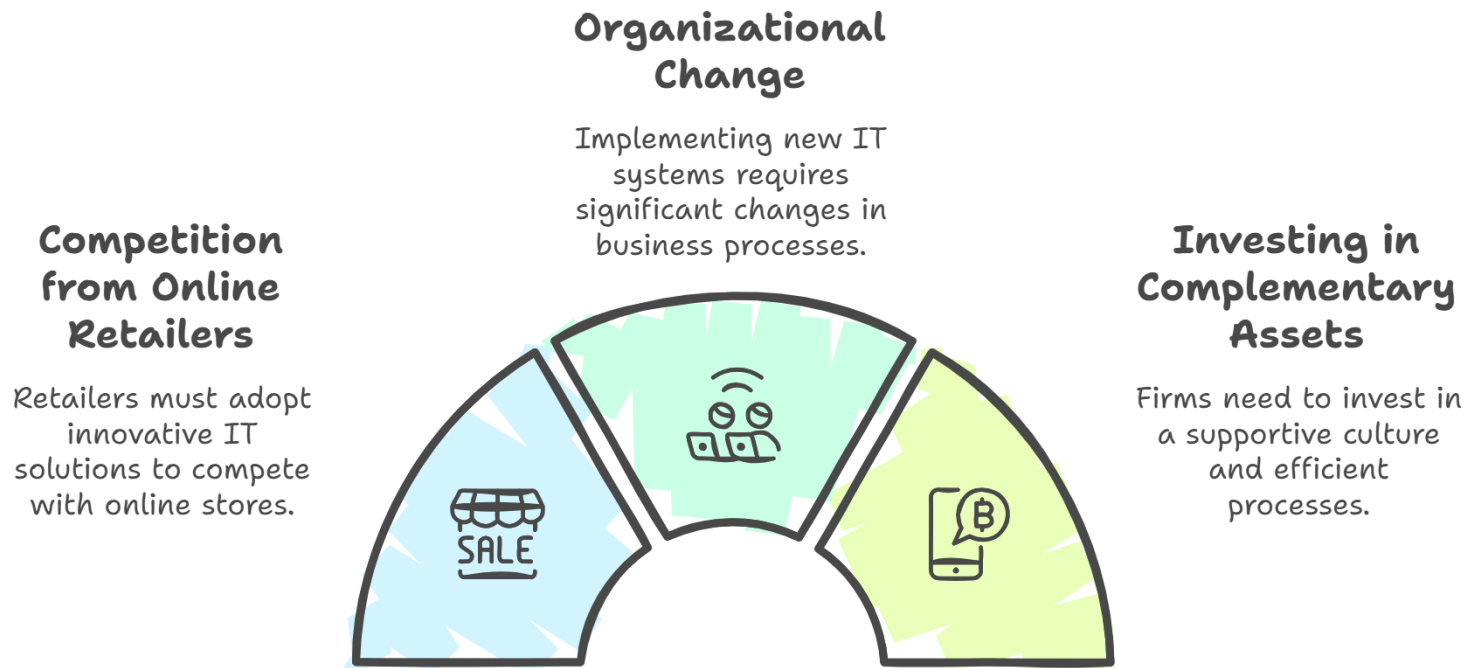




Example: Retail Stores

Some Challenges faced by Retail Stores

Competitiveness and Efficiency Challenges in Retail Store



Need to invest in Complementary assets

- **Technology alone is not enough**
 - success depends on complementary investments.
- Retailers must build: A supportive organizational culture
- Staff training programs to boost adoption and reduce resistance
- Streamlined processes that maximize the value of new technologies
- **Example:**
 - Central Sterilization Services Department (CSSD) at a local hospital adopted RFID for inventory tracking
 - Success driven by process reengineering and targeted staff training



► J Clean Prod. 2014 May 4;85:382–394. doi: [10.1016/j.jclepro.2014.04.037](https://doi.org/10.1016/j.jclepro.2014.04.037)

RFID-enabled process reengineering of closed-loop supply chains in the healthcare industry of Singapore

[Arun Kumar](#)^{a,1}, [Shams Rahman](#)^{b,*}

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PMCID: PMC7127495 PMID: [32288347](#)

Abstract

Due to an increasing global awareness of the need for sustainable development in academia and business, closed-loop supply chains in hospitals have become an important area of focus. Process reengineering in purchasing, materials management, work practices, and waste management can reduce environmental hazards and contain costs at the healthcare facilities. This paper demonstrates an application of the RFID-enabled process reengineering in sustainable healthcare system design, and presents a case study in the linens division of central sterilization services department at a Singaporean hospital using ARENA simulation.

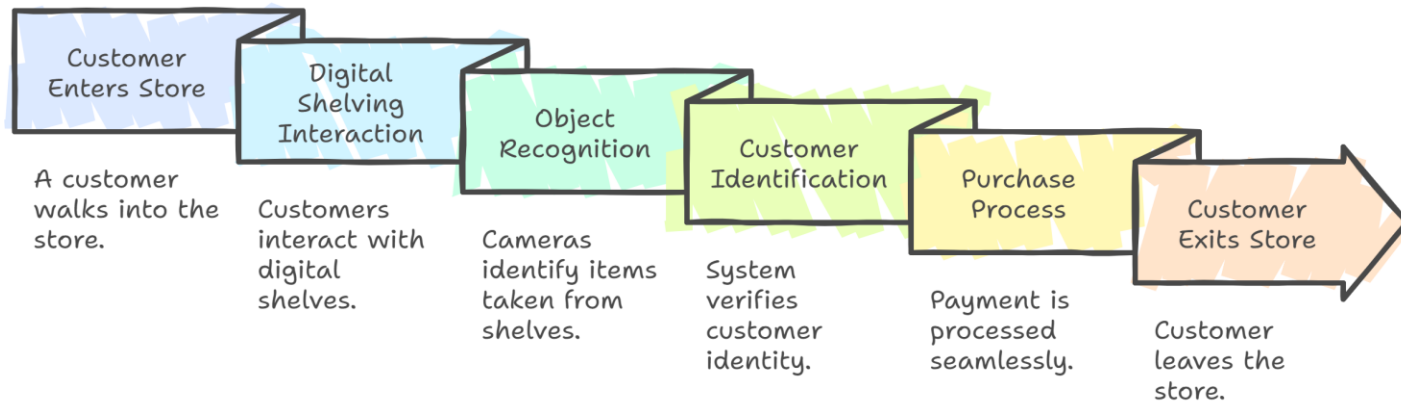
Keywords: Closed-loop supply chain, Healthcare system, Linens department, Process reengineering, Singapore

[source](#)

IS enabled Enhancements

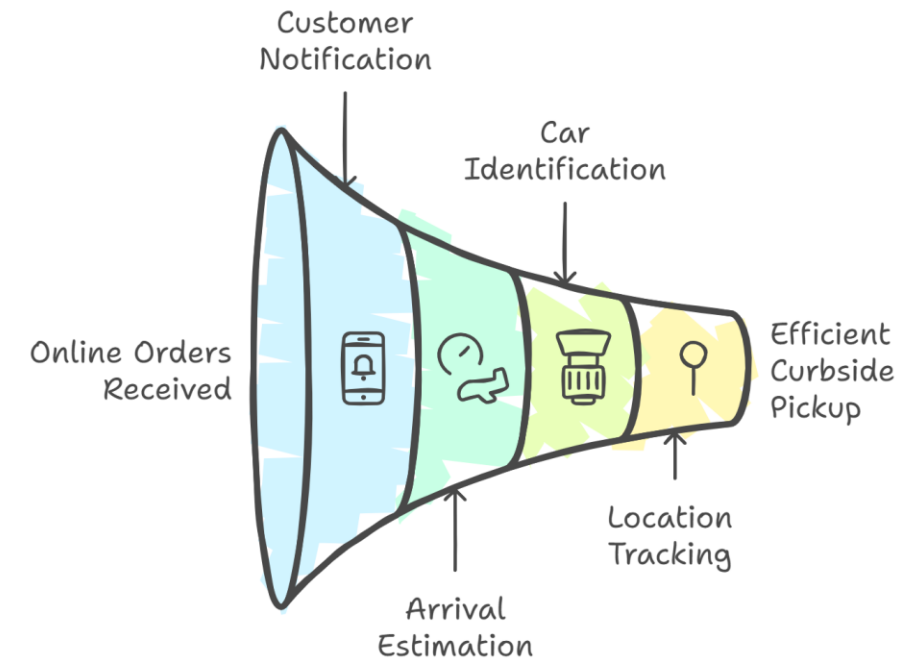
- Helps in-house customers and
- Can streamline product pickups

AWM Frictionless Shopping Experience

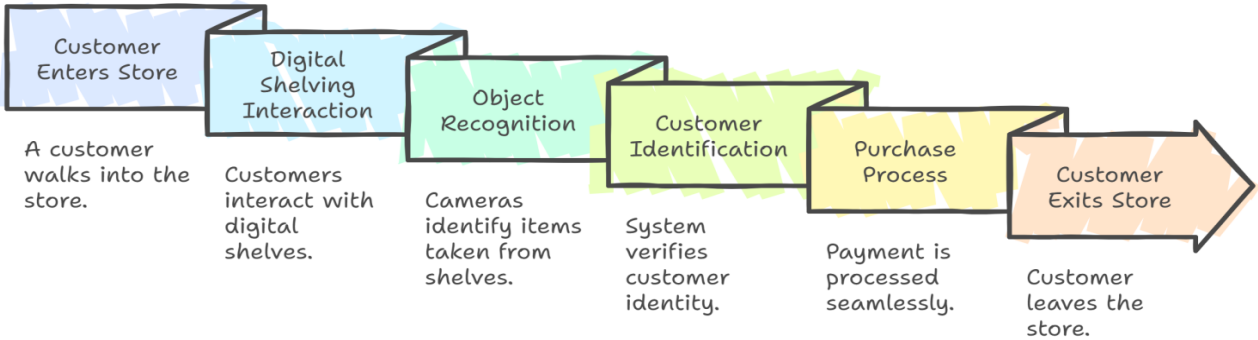


[Watch a video on AWM](#)

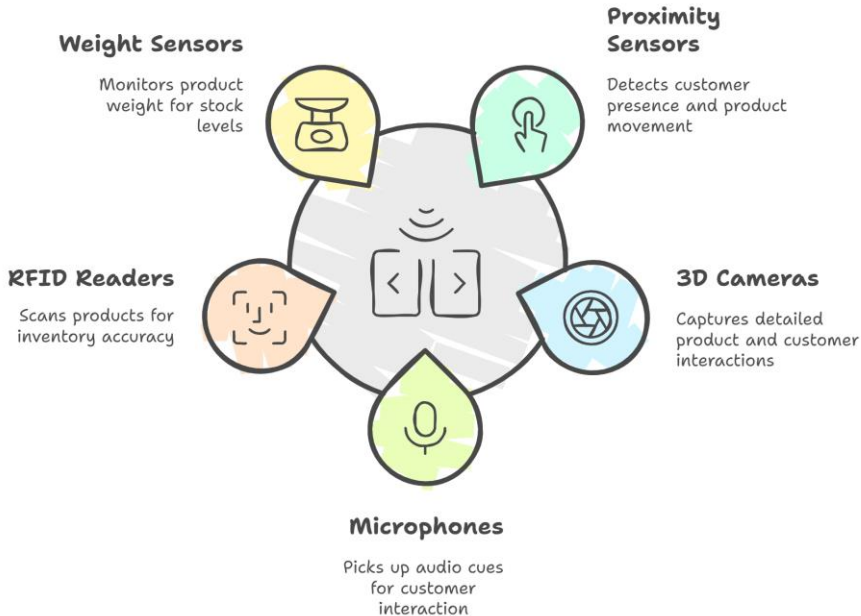
Streamlined Curbside Pickup Process



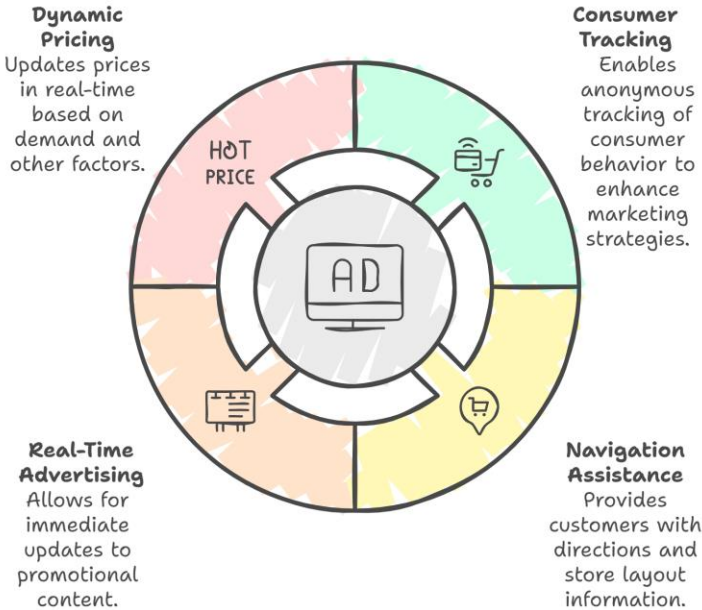
AWM Frictionless Shopping Experience



AWM Smart Shelf Technology

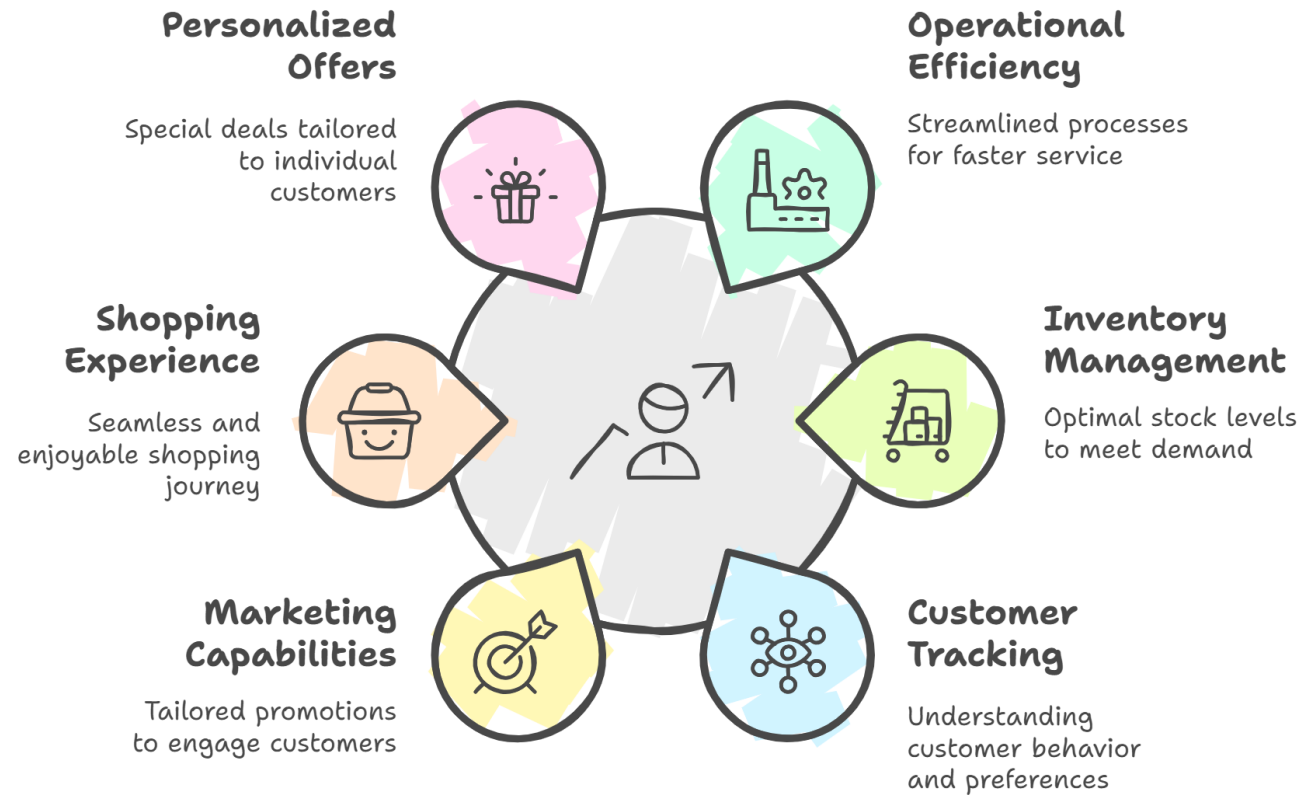


Digital Signage Applications



IS enabled Enhancements

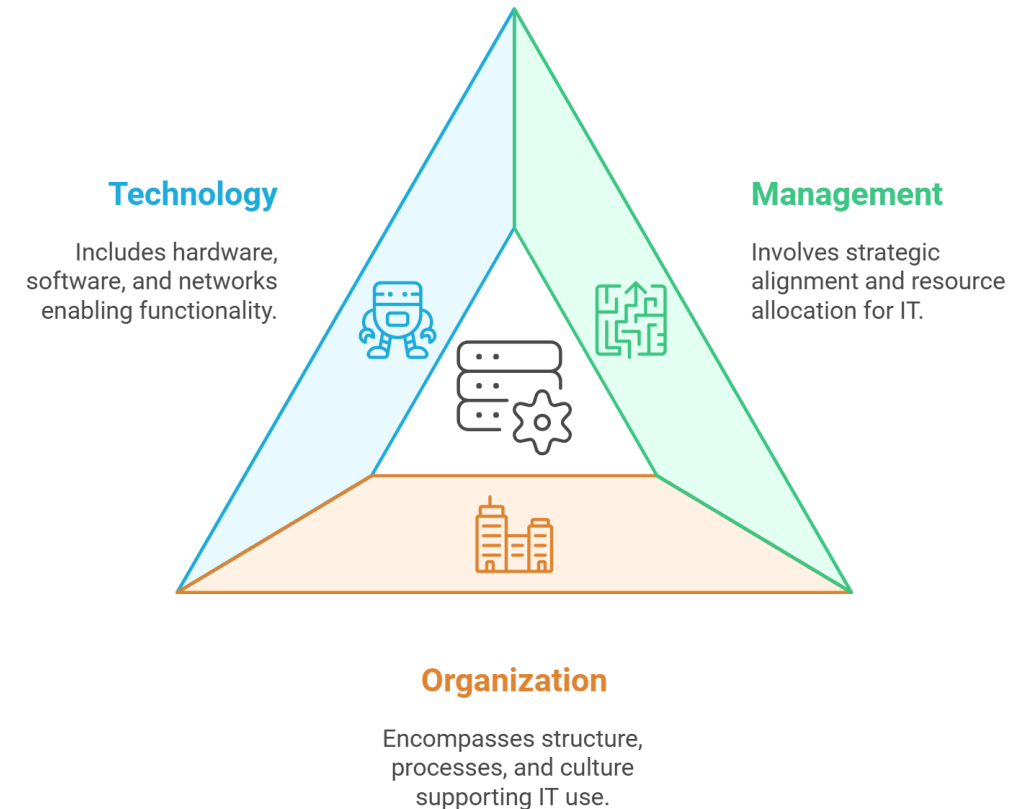
Key Benefits of Retail Enhancements



IS System: Definition

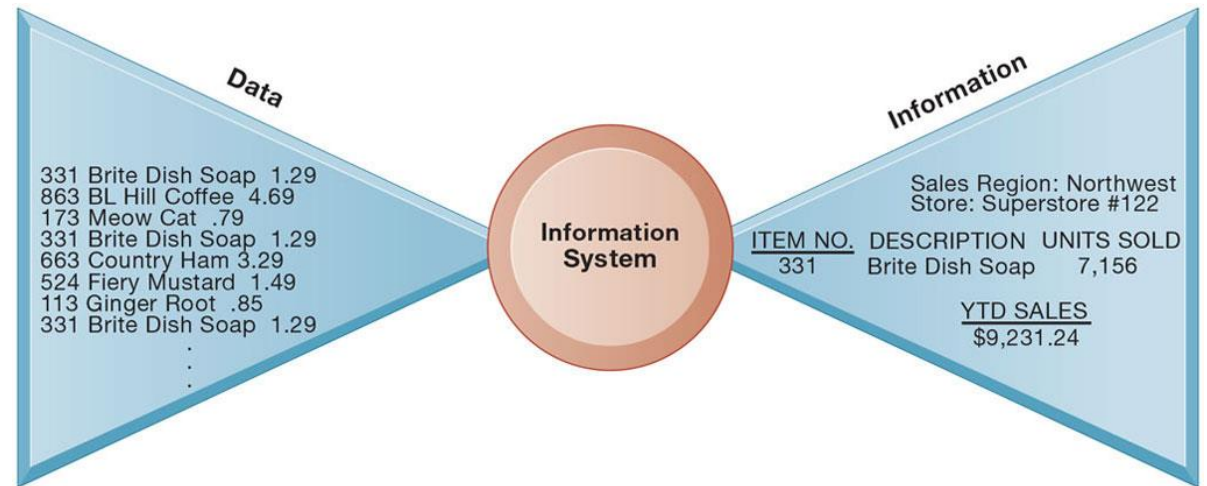
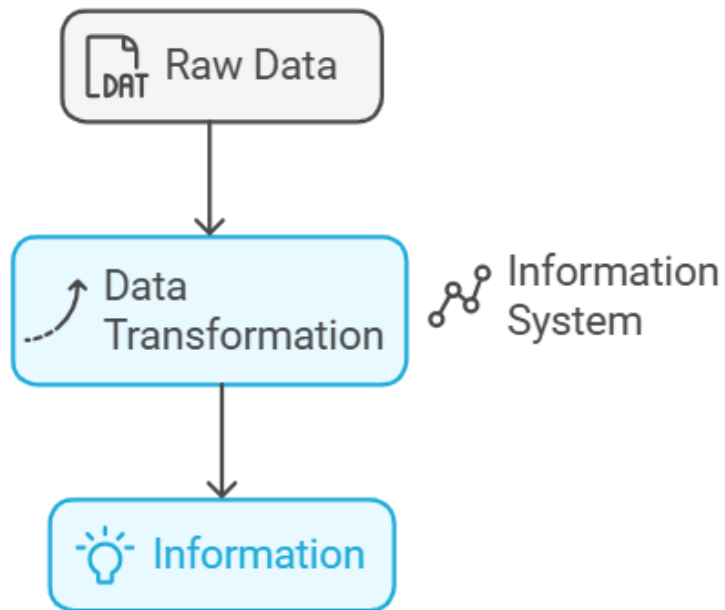
- Information systems
 - Set of interrelated components
 - Collect, process, store, and distribute information
- Support:
 - ✓ Decision-making
 - ✓ Coordination
 - ✓ Control
 - ✓ Analysis
 - ✓ Visualisation

Key Components of An Information System



IS System: Data to Information

- Information vs. Data: Information is data shaped into meaningful form, a representation of reality

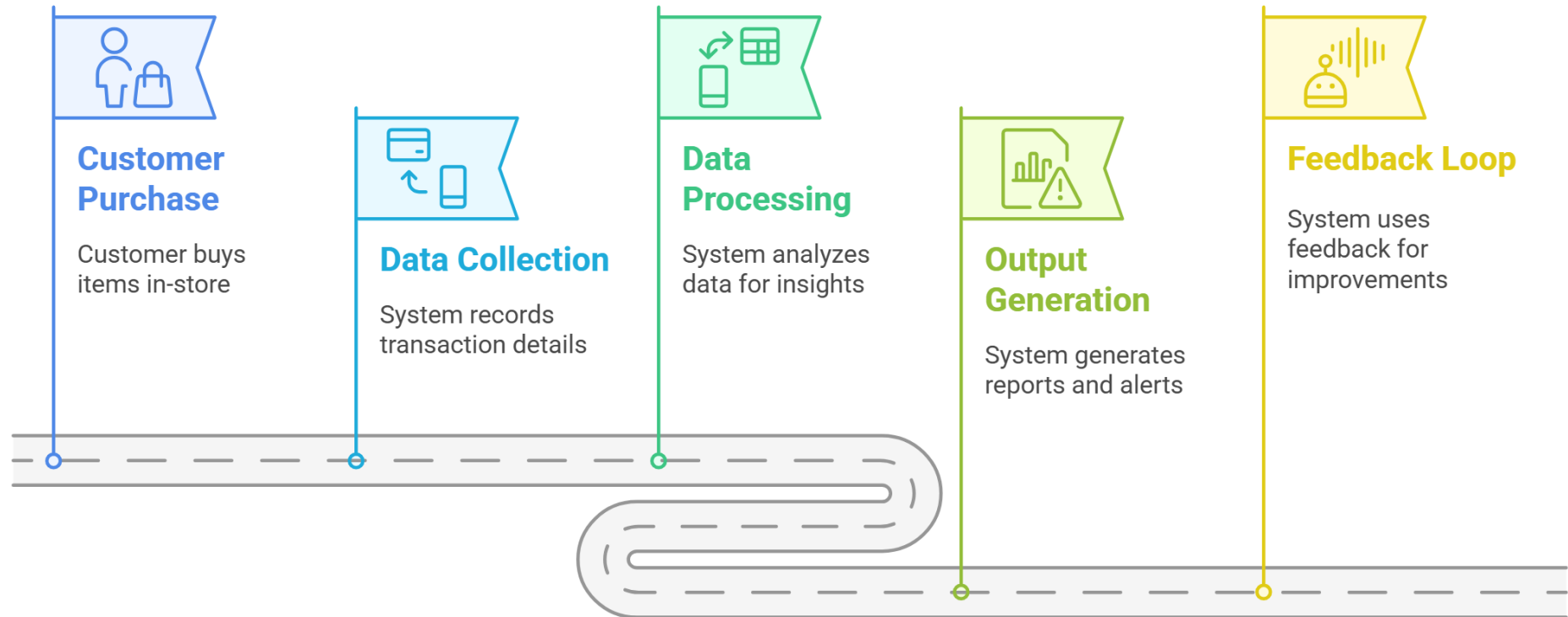


IS System: Data to Information

- Activities of information systems that produce information
 1. Input
 - Captures raw data from organization or external environment
 2. Processing
 - Converts raw data into meaningful form
 3. Output
 - Transfers processed information to people or activities that use it
 4. Feedback
 - Output is returned to appropriate members of organization to help evaluate or correct input stage
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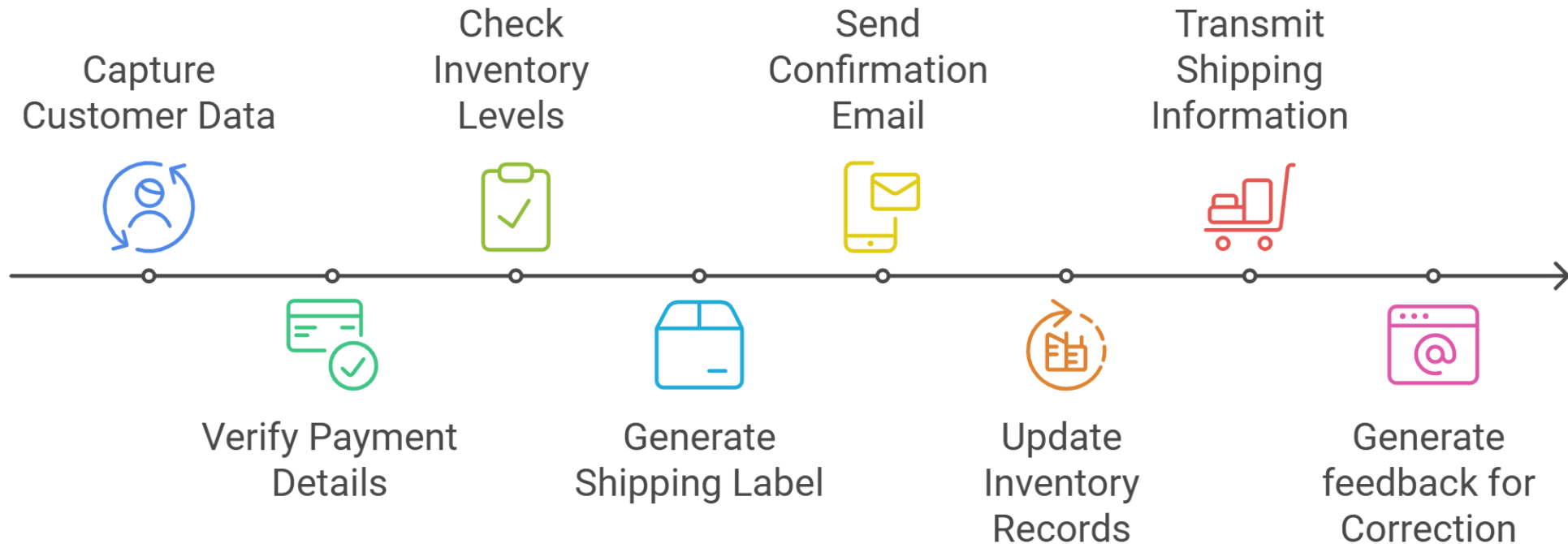
IS System: Data to Information (example)

Retail Transaction Data Flow



IS System: Data to Information (example)

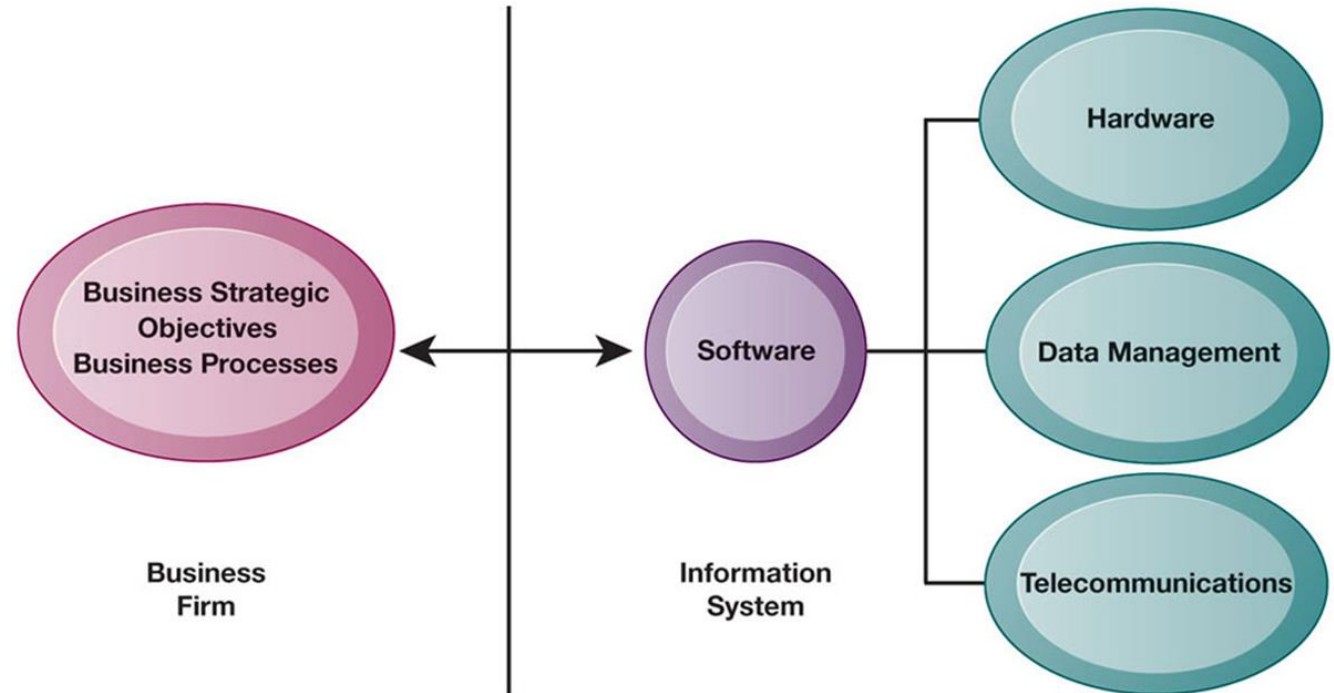
Example: Online Retail System Process



Business-IS Interdependence

Growing interdependence between:

- Ability to use information technology
- Ability to implement corporate strategies and achieve corporate goals



Business-IS Interdependence

Modern Business Challenges

- Need for real-time decisions
- Demand for digital transformation
- Competitive pressure to innovate
- Required system flexibility

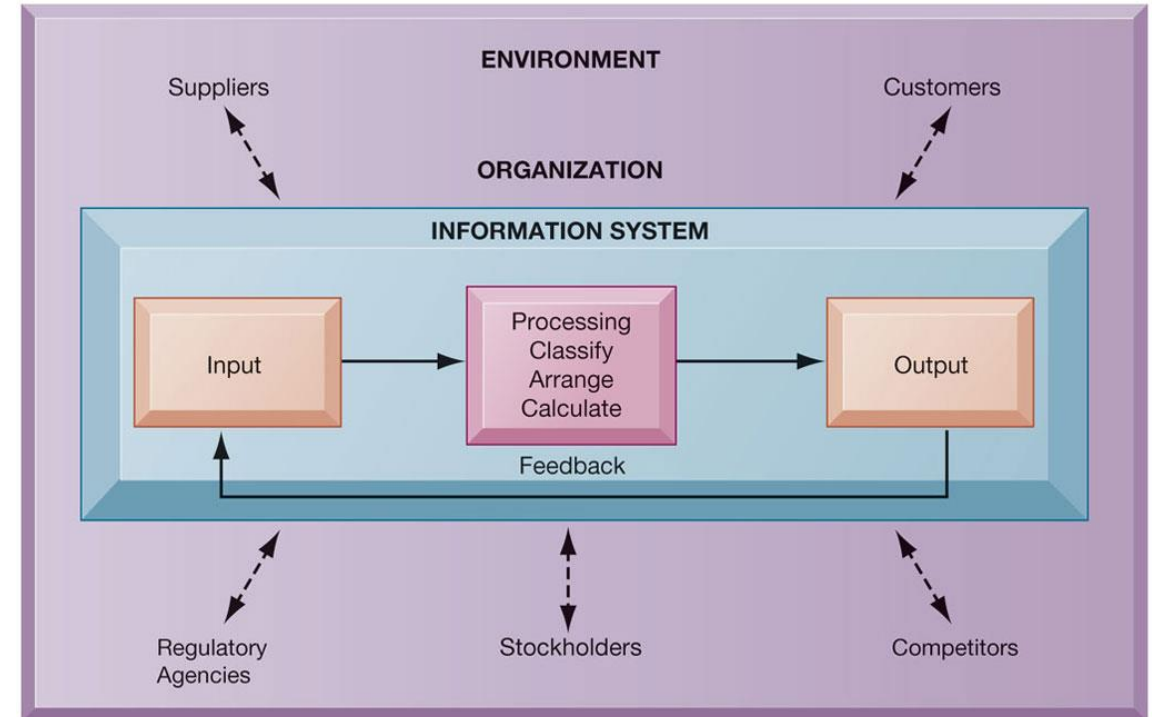
Information Systems Shape Business Capabilities

- Business strategies depend on IT capabilities
 - Changes in business require changes in IT
 - IT limitations can restrict business options
 - IT innovations enable new business opportunities for all stakeholders
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IS System: Key Stakeholders

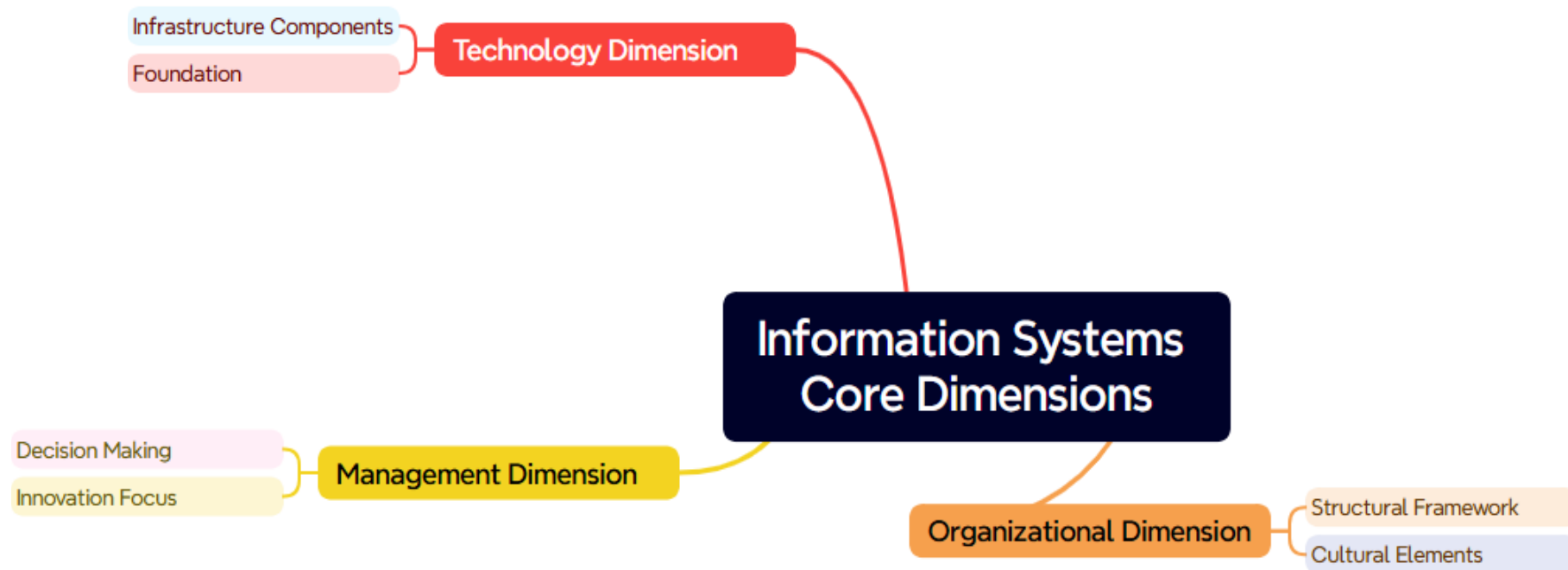
The organization's information system operates within a broader environment, interacting with key stakeholders:

- **Customers**, and **Suppliers** (business relationships),
- **Stockholders**, and **Regulatory agencies** (governance/compliance),
- **Competitors** (market dynamics)
- This interconnected structure ensures the organization can effectively collect, process, and utilize information to meet business needs while responding to environmental factors



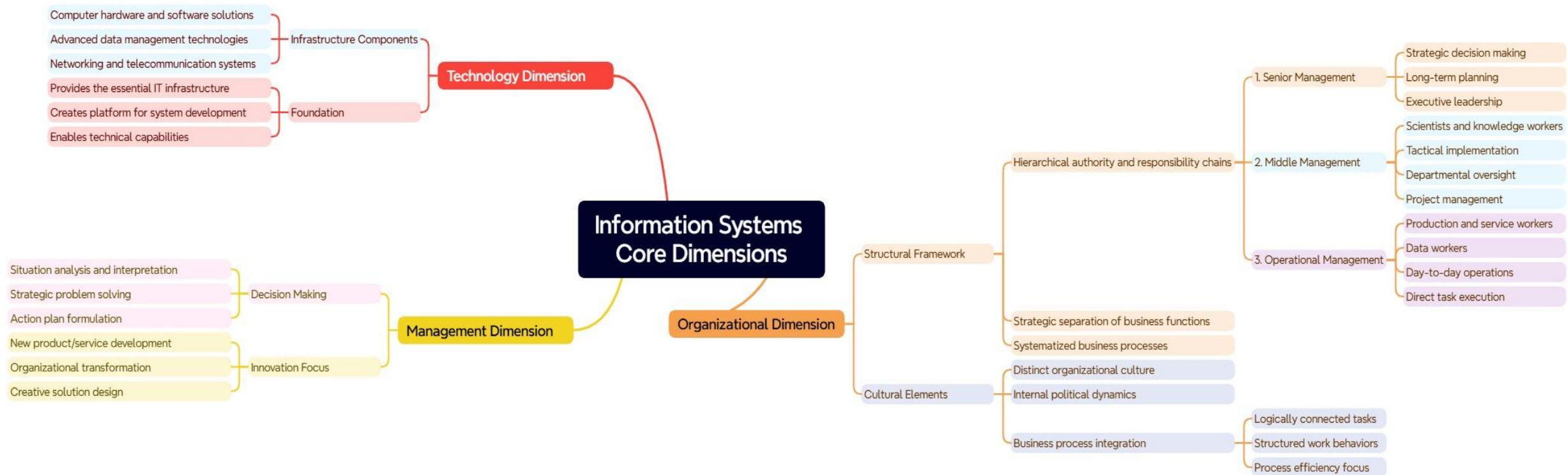
IS System: Dimensions

At the centre, Information Systems serve as the convergence point where all three dimensions meet and interact, creating a comprehensive framework for modern business operations.

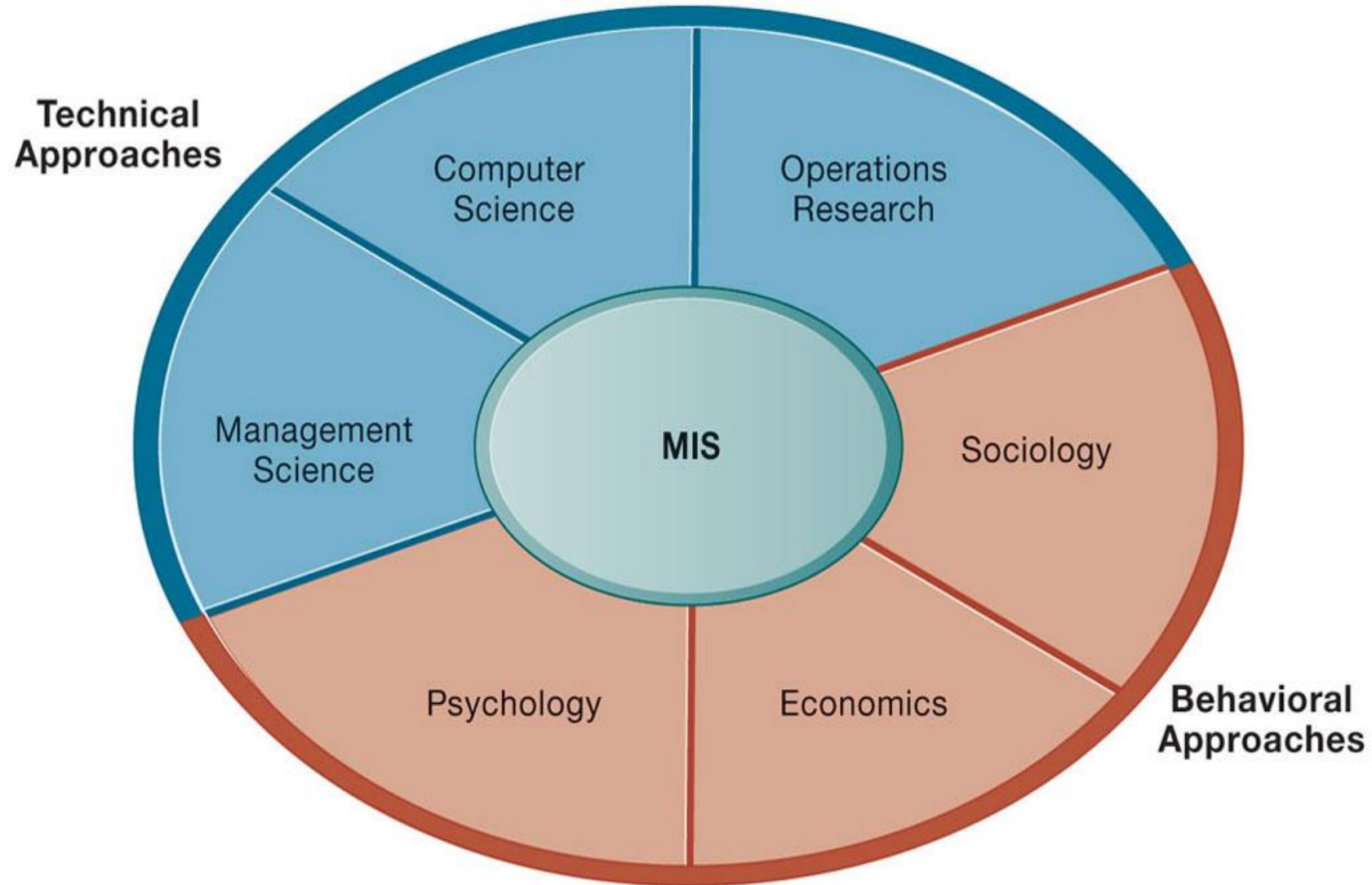


Note: Each dimension is equally critical for the successful implementation and operation of information systems within an organization.

IS System: Dimensions (contd.)



Study an IS System



Study an IS System: Core Approaches

- 3 Core Approaches

- Technical Approach

- Computer Science (Hardware, Software, Networks)
- Management Science (Process Optimization)

- Behavioural Approach

- Sociology & Psychology
- Economics & Political Science

- Sociotechnical view

- Integration of Social + Technical Systems
- Helps avoid purely technological approach

- Supporting Fields

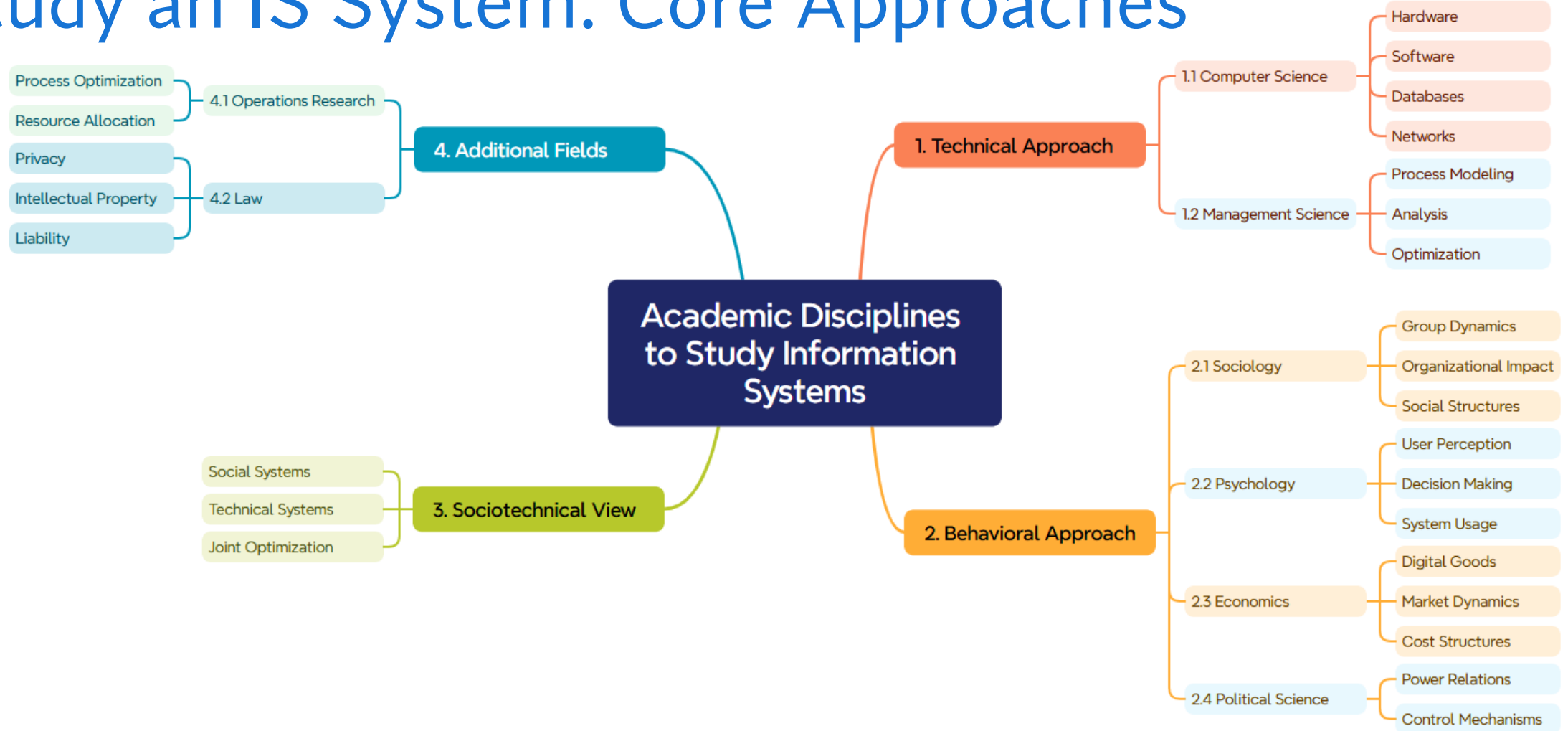
- ☐ Operations Research

- Process Optimization
- Resource Allocation

- ☐ Legal Studies

- Privacy
 - Intellectual Property
-

Study an IS System: Core Approaches



Key Points Covered (1/3)

- What is an Information System (IS)?
 - The IS Process: Data to Information involves four key stages:
 - Input: Capturing raw data from internal and external sources.
 - Processing: Converting raw data into a useful and meaningful format.
 - Output: Transferring the processed information to users who need it within the organization.
 - Feedback: Returning output to relevant members of the organization to evaluate and correct the input process.
 - Core Dimensions of an Information System
 - Technology Dimension
 - Organisational Dimension
 - Management Dimension
-

Key Points Covered (2/3)

- Business-IS Interdependence
 - Key Stakeholders
 - Business Relationships: Customers and Suppliers
 - Governance/Compliance: Stockholders and Regulatory agencies
 - Market Dynamics: Competitors
 - This interconnectedness ensures the organization can effectively use information to meet its goals while adapting to external influences.
 - Academic Disciplines in IS Study
 - Technical Approach
 - Computer Science: Focuses on the hardware, software, databases, and networks that form the technical infrastructure.
 - Management Science: Emphasizes process modelling, analysis, and optimization.
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Key Points Covered (3/3)

- Academic Disciplines in IS Study
 - Behavioral Approach:
 - Sociology: Explores group dynamics, organizational impact, and social structures.
 - Psychology: Concentrates on user perception, decision-making, and system usage.
 - Economics: Examines digital goods, market dynamics, and cost structures.
 - Political Science: Considers power relations and control mechanisms in relation to IS.
 - Sociotechnical View: This approach recognizes that information systems are composed of both social and technical elements, and that they must be jointly optimized for success.
 - Additionally, there are supporting fields:
 - Operations Research: Process optimization and resource allocation.
 - Law: Privacy, intellectual property, and liability considerations.
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Class Activity

Identify Key Components of IT in Organizations

- Homework Activity
 - doc: L1_Activity.pdf



This Week's action items for you

- A. Sign-up on Canvas for your project
- B. Designate a team leader (Team leader to communicate on team's behalf)

